

Machine-verified rooted-tree majorants for polymer expansions with holes

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Abstract

We present a machine-verified quantitative toolkit for cluster expansions of polymer systems with excluded regions (*holes*), in the discrete cube geometry of Balaban–Dimock renormalization-group analyses. Five Lean 4 theorems, compiled against a pinned Mathlib revision with a clean axiom oracle, provide: (i) the *exact identity* $\sum_T \prod_v c_T(v)! = n! C_n$ for child factorials over spanning trees of the complete graph K_{n+1} , with the rooted-tree majorant 4^n as corollary; (ii) a marked-root leaf summation for the tree-graph majorant of the second Ursell expansion with holes, with the moment constant M paid once at the root and closed leaf ratio $4M^2$ per additional vertex, together with its Catalan-sharpened form $M^{2n+1}C_n$ (a gain of order $n^{3/2}$ at order n); and (iii) a target-preserving orderwise bound in which the exact target union survives until the modified-metric exponential is extracted. A certified companion (interval arithmetic, 120-bit precision, committed transcript, reproduced from scratch) tabulates the smallness gate and encloses every derived constant. Non-vacuity is machine-checked: a concrete hole family satisfying every hypothesis is exhibited in Lean and both polymer-facing endpoints are instantiated at it with a strictly positive weight. The paper states what is proved, why it is true, and exactly which layer each claim lives in: Lean theorem, interval certificate, or paper-level algebra.

1 Introduction

Cluster (polymer) expansions convert log-partition functions into sums over connected geometric objects; their convergence proofs hinge on combinatorial majorants for tree sums and on geometric smallness gates [1, 2, 3]. When the expansion must respect *holes* — excluded regions that the expansion sees as atomic units, as in renormalization-group treatments of large-field regions [6, 8] — the bookkeeping is notoriously error-prone: root normalizations, child factorials, and hole-separation hypotheses interact in ways that invite wrong constants.¹

This paper contributes, in machine-checked form:

1. an exact Catalan evaluation of the rooted child-factorial tree sum (Theorem 3.1), of which the classical 4^n tree-shape budget (Theorem 3.2) is a corollary;

*With machine collaboration; every theorem below is machine-checked in Lean 4 against a pinned Mathlib, and every numerical constant is enclosed by committed interval arithmetic. Repository: <https://github.com/lluiseiriksson/THE-ERIKSSON-PROGRAMME>.

¹Our own development measured exactly such an error in its pre-registered prediction of the smallness-gate crossover; see Remark 5.1. The correction protocol — constants are computed, never assembled by hand — is part of the method.

2. finite, volume-free bounds for the marked-root and fixed-target-union tree-graph majorants of a second Ursell expansion with holes (Theorems 3.3, 3.4, 3.5), with every constant derived from one smallness gate;
3. a certified constant table (Section 5) and a machine-checked non-vacuity witness (Section 7).

Statuses are kept separate throughout: **exact** means a Lean theorem in the pinned core with axiom oracle `[propext, Classical.choice, Quot.sound]`; **certified** means a committed interval-arithmetic transcript with ball-plus-boolean output; **paper-level** means ordinary mathematical prose not formalized here, and is always flagged. No continuum or Yang–Mills claim of any kind appears: importance is not inherited by thematic proximity, and no proved reduction to any such problem is claimed.

[SLOT: oracle output of the five endpoints and two non-vacuity theorems post-build; axiom list must print exactly `[propext, Classical.choice, Quot.sound]`]

2 Setting and definitions

Throughout, $d, L \in \mathbb{N}$ with $L \neq 0$.

Definition 2.1 (cubes and adjacency). *A cube is a point of the discrete torus $(\mathbb{Z}/L)^d$ (Lean: `Cube d L`). Two cubes are adjacent (`cubeAdj`) iff they are distinct and each coordinate differs by 0 or ± 1 ; this is the sup-metric neighbor graph, in which every cube has $3^d - 1$ neighbors. A cube set is connected when it is walk-connected in this graph.*

Definition 2.2 (hole families and skeletons). *A hole family $\mathcal{H}$ (Lean: `HoleFamily d L`) is a finite set of finite cube sets — the connected components of the excluded region. A cube set X respects $\mathcal{H}$ iff every hole $H \in \mathcal{H}$ satisfies $H \subseteq X$ or $H \cap X = \emptyset$: the expansion sees each hole as an atomic unit [8, eq. (149)]. The skeleton $\text{sk}(X)$ is the set of cubes of X lying in no hole. The bounds below hypothesize: pairwise disjointness of holes, no adjacency edge between distinct holes, and non-emptiness of each hole.*

Definition 2.3 (polymers). *Given $\mathcal{H}$ and an activity $z: \mathcal{P}((\mathbb{Z}/L)^d)_{\text{fin}} \rightarrow \mathbb{C}$, a polymer is a nonempty, connected, hole-respecting cube set with nonempty skeleton; Ω denotes the (finite) type of polymers (Lean: `OmegaPolymerType`). Two polymers are incompatible, written $X \approx X'$, iff their skeletons intersect. For a tuple $\mathbf{X} = (X_0, \dots, X_n) \in \Omega^{n+1}$, its incompatibility graph $G(\mathbf{X})$ has vertex set $\{0, \dots, n\}$ and an edge $\{i, j\}$ iff $X_i \approx X_j$.*

Definition 2.4 (modified metric and hole weight). *The discrete modified metric $\rho(X)$ (Lean: `discreteModifiedMetric`) is the Steiner length of the skeleton inside X :*

$$\rho(X) = \min\{|S| - 1 : \text{sk}(X) \subseteq S \subseteq X, S \text{ connected}\},$$

with $\rho(X) = 0$ if no such S exists. For a rate $\kappa \geq 0$ the hole weight is $W_\kappa(X) = e^{-\kappa(\rho(X)+1)}$ (Lean: `appendixFHoleExpWeight`).

Definition 2.5 (the two tree-graph majorants). *Fix $\mathcal{H}$, an activity z , a nonnegative weight $w: \Omega \rightarrow \mathbb{R}_{\geq 0}$, and $n \in \mathbb{N}$. Write $\tau(\mathbf{X})$ for the number of spanning trees of $G(\mathbf{X})$. The marked-root sum at a cube r is*

$$S_{w,r}^\#(n) = \frac{1}{(n+1)!} \sum_{\substack{\mathbf{X} \in \Omega^{n+1} \\ r \in \text{sk}(X_0)}} \tau(\mathbf{X}) \prod_{j=0}^n w(X_j)$$

(Lean: `appendixFHoleHsharpWeightedTreeMarkedRootSum`), and the fixed-target sum at a finite cube set Y is

$$T_{w,Y}^{\#}(n) = \frac{1}{(n+1)!} \sum_{\substack{\mathbf{X} \in \Omega^{n+1} \\ \bigcup_j X_j = Y}} \tau(\mathbf{X}) \prod_{j=0}^n w(X_j)$$

(Lean: `appendixFHoleHsharpWeightedTreeTerm`). These are the standard tree-graph majorants of the order- $(n+1)$ coefficients of a second Ursell (Mayer) expansion: by tree-graph inequalities of Penrose type [4, 5], Ursell coefficients are dominated by spanning-tree counts of incompatibility graphs, and it is the majorant objects $S^{\#}, T^{\#}$ — not the signed coefficients — that the theorems below bound. The passage from signed coefficients to these majorants lives upstream in the repository’s Kotecký–Preiss/Mayer–Ursell development and is not re-proved here.

Definition 2.6 (gate and derived constants). The smallness gate at dimension d and rate $\kappa_0 > 0$ is

$$G(d, \kappa_0) = (3^d)^2 e^{-\kappa_0} 2^{3^d+1} < 1 \quad (\text{Lean hypothesis } h_{C_q}), \quad (1)$$

and the derived constants are

$$\text{RS} = (1 - G)^{-1}, \quad M = \max(1, \kappa_0^{-1}, \text{RS}), \quad \Lambda = 4M^2 \quad (2)$$

(Lean: `appendixFSecondUrsellMomentConstant`, `appendixFSecondUrsellLeafConstant`). In applications where the order- n term additionally carries an activity factor ε^n , the resulting geometric series has ratio $\Lambda\varepsilon$, which is below one exactly for $\varepsilon < \varepsilon^* := \Lambda^{-1}$ (strict; nothing is claimed at equality). Status: the Lean endpoints prove the coefficient bounds $M\Lambda^n$ and $M^{2n+1}C_n$ (exact); the enclosures of Λ^{-1} are certified; the passage from coefficient bound to a convergent activity series is paper-level algebra conditional on the stated ε^n structure.

3 Main results

All five statements are stable aliases in module `MarkedRootedClosure`, file `PaperTheorems.lean`; each is an exact application of a pinned upstream theorem — no new axiom, no opaque assumption. Here $\mathcal{T}(K_{n+1})$ is the set of spanning trees of the complete graph on the vertices $\{0, \dots, n\}$, each tree rooted at 0, and $c_T(v)$ is the number of children of v in the rooted tree T .

Theorem 3.1 (exact Catalan value; `normalizedRootedChildFactorialTreeCatalanIdentity`). For every $n \in \mathbb{N}$,

$$\frac{n+1}{(n+1)!} \sum_{T \in \mathcal{T}(K_{n+1})} \prod_v (c_T(v))! = C_n, \quad \text{equivalently} \quad \sum_T \prod_v c_T(v)! = n! C_n,$$

where $C_n = \binom{2n}{n}/(n+1)$ is the n th Catalan number.

Theorem 3.2 (rooted-tree majorant; `normalizedRootedChildFactorialTreeBound`). For every n , the left-hand side of Theorem 3.1 is at most 4^n .

Theorem 3.2 follows from Theorem 3.1 via $C_n \leq 4^n$; the classical budget overcounts by exactly $4^n/C_n \sim \sqrt{\pi} n^{3/2}$.

Theorem 3.3 (marked-root leaf summation; `markedRootLeafGeometricBound`). *Let $\mathcal{H}$ satisfy the separation hypotheses of Definition 2.2, let $w \geq 0$ be dominated by the hole weight at doubled rate, $w \leq W_{2\kappa_0}$ pointwise, let r be any cube, and assume the gate (1). Then for every n ,*

$$(n+1) \cdot S_{w,r}^\#(n) \leq M \cdot \Lambda^n.$$

Theorem 3.4 (Catalan-sharpened marked-root bound; `markedRootCatalanBound`). *Under exactly the hypotheses of Theorem 3.3,*

$$(n+1) \cdot S_{w,r}^\#(n) \leq M^{2n+1} \cdot C_n.$$

Since $M\Lambda^n = M^{2n+1}4^n$, this improves Theorem 3.3 by the factor $4^n/C_n$: the tree-shape factor is paid at its exact Catalan value.

Theorem 3.5 (target-preserving orderwise bound; `targetPreservingWeightedTreeBound`). *With the same geometric hypotheses, let $w \leq W_{\text{rate}} \cdot u$ be a pointwise split of the weight into an extracted exponential at a rate ≥ 0 times a remainder $u \leq W_{2\kappa_0}$, and let Y be a finite cube set whose skeleton contains some cube r . Then*

$$T_{w,Y}^\#(n) \leq M \cdot W_{\text{rate}}(Y) \cdot \Lambda^n.$$

4 Proof architecture

The formal proofs are the authority; this section gives the mathematical mechanism at a level where a specialist can reconstruct the logic. Module pointers are to the repository.

4.1 The Catalan identity

The sum $\sum_T \prod_v c_T(v)!$ counts pairs (T, σ) where T is a spanning tree of K_{n+1} rooted at 0 and σ assigns to each vertex a linear order on its children — i.e., it counts *labelled rooted plane trees* on the vertex set $\{0, \dots, n\}$ with root 0. Plane trees are rigid (a plane structure admits no nontrivial automorphism fixing the root), so this count factors as

$$\#\{\text{plane shapes with } n+1 \text{ nodes}\} \times \#\{\text{labellings of the } n \text{ non-root nodes}\} = C_n \cdot n!,$$

the first factor by the classical bijection between plane trees and binary trees, formalized in `YangMills/KP/PlaneTreeCatalan.lean` through Mathlib’s

```
Tree.treesOfNumNodesEq_card_eq_catalan;
```

the identity then normalizes to Theorem 3.1 (`YangMills/KP/RootedCatalanExact.lean`).

Example 4.1. $n = 2$: K_3 has three spanning trees. Rooted at 0, the two paths through 0 contribute $\prod_v c_T(v)! = 1$ each and the star at 0 contributes $2! = 2$; the sum is $4 = 2! C_2$.

4.2 Leaf summation and the $2n + 1$ moment budget

Theorems 3.3 and 3.4 are proved by splitting the tuple sum by the (rooted) shape of a spanning tree and peeling leaves. Three mechanisms produce the constants:

One-pin polymer sums. The basic estimate controls the sum of $W_{2\kappa_0}(X)$ over polymers X whose skeleton contains a fixed cube. A polymer with $\rho(X) + 1 = m$ has a skeleton of at most m cubes carried by a connected Steiner set; enumerating connected supersets cube by cube costs the local entropy 2^{3^d+1} per step and the pin transfer between neighboring cubes costs $(3^d)^2$, so the sum over polymers at Steiner length m is dominated by G^m with G the gate (1), and the total by the geometric sum $RS = (1 - G)^{-1} \leq M$. This is the only place the gate enters, and it is why a single inequality controls every constant.

Moment paid once at the root. In the marked-root sum the coordinate X_0 is pinned at r , so the root contributes one one-pin sum: one factor M . Each tree edge propagates a pin: a child polymer must have skeleton intersecting its parent's, so summing the child costs one one-pin sum (M) after choosing the pin inside the parent, whose cost is accounted to the parent's own moment. The bookkeeping closes with each of the n non-root vertices paying two factors (M for its own one-pin sum, M for the pin it offers its children) and the root paying one: M^{2n+1} total. The formal count is the fixed-parent kernel lemma

```
appendixFHoleHsharpWeightedTreeMarkedRootFixedParentKernelSum
    _le_prod_childMomentBudget_mul_rootMoment
```

and its consumers in `YangMills/RG/AppendixFSecondUrsellLeafSummation.lean`.

Shape sum. What remains is the sum over rooted tree shapes with child multiplicities — exactly the object of Section 4.1. Paying it at its Catalan value gives Theorem 3.4; paying it at the 4^n budget and absorbing $4^n M^{2n} = \Lambda^n$ gives Theorem 3.3. The $(n + 1)$ prefactor in both statements is the marked-coordinate count: pinning coordinate 0 rather than an arbitrary coordinate costs a factor $n + 1$ against the $(n + 1)!$ normalization.

4.3 Target preservation

In Theorem 3.5 the tuple union is constrained to a fixed target Y . Extracting the metric exponential *before* summing would discard this constraint and with it the geometry of Y . Instead, the hypothesis $w \leq W_{\text{rate}} \cdot u$ is applied per polymer, and the extracted factors combine across any tuple with union Y into at least the full-target factor $W_{\text{rate}}(Y)$ — a superadditivity step for the Steiner length over coverings. Only after this exact factor is secured is the remainder u routed through the marked-root bound at a root $r \in \text{sk}(Y)$. The order of operations is the content of the theorem's name and the reason the constant is volume-free in Y .

5 The certified companion

Script `scripts/c1_admissibility_arb.py` (SHA-256

```
f114fc4bcf2193c47603fb595b91b4e3ea62addc3a1bfebe0aa0c147c447f1f9
```

`python-flint` 0.9.0, precision 120 bits; transcript committed alongside it, exit 0) encloses G , M , Λ , ε^* with ball-plus-boolean output; the gate booleans are provable-only (PASS printed only when $G < 1$ holds for every point of the enclosing balls). The full run was re-executed from scratch during final assembly and reproduced the committed transcript byte-identically. Selected rows (midpoints shown; radii in the transcript, all below 3×10^{-6} relative):

d	κ_0	gate G	M	Λ	ε^*
2	11	1.3853059	FAIL (provably ≥ 1)		
2	11.5	0.84023048	6.2590162	156.70114	0.0063815746
2	12	0.50962555	2.0392580	16.634292	0.060116775
2	16	0.0093341175	1.0094221	4.0757316	0.24535472
2	24	3.13×10^{-6}	1.0000031	4.0000251	0.24999843
3	25	2.7177241	FAIL (provably ≥ 1)		
3	26	0.99979481	4873.5020	9.50×10^7	1.05×10^{-8}
3	28	0.13530751	1.1564805	5.3497886	0.18692327
3	33	0.00091169486	1.0009125	4.0073035	0.24954436

The certified crossovers of the gate are

$$\kappa^*(2) \in 11.32592096 \pm 2.72 \times 10^{-10}, \quad \kappa^*(3) \in 25.99979479 \pm 2.32 \times 10^{-9},$$

and $\varepsilon^* \uparrow 1/4$ as $\kappa_0 \rightarrow \infty$ (the pure tree-majorant limit $\Lambda \downarrow 4$). The row $(d, \kappa_0) = (3, 26)$ sits just above $\kappa^*(3)$ and displays the sharpness of the gate: admissible, but with $M \approx 4.9 \times 10^3$ and $\varepsilon^* \approx 1.05 \times 10^{-8}$.

Scope of the certificates: the transcript’s band booleans certify the gate at the two registered band endpoints $\kappa_0 = 16$ and $\kappa_0 = 33$; the extension to all larger κ_0 is the exact monotonicity of $\kappa_0 \mapsto e^{-\kappa_0}$ (one line, not part of the certified output). Admissibility everywhere means the strict gate $G < 1$, as in the Lean hypothesis.

Remark 5.1 (a measured registration failure). *The pre-registered charter predicted crossovers 15.72 and 32.60 from a hand-assembled expression carrying $(3^d)^4$ where the gate (1) carries $(3^d)^2$. The certified run measured the discrepancy (transcript, “crossover audit”: REGISTERED PREDICTION DOES NOT RE-DERIVE); the registered admissibility bands $\kappa_0 \geq 16$ ($d=2$) and $\kappa_0 \geq 33$ ($d=3$) remain true — sufficient, not sharp — and only certified values are inked here. We record the failure because the method requires it: constants are computed, never assembled.*

6 What the Catalan sharpening buys

The gain of Theorem 3.4 over Theorem 3.3 at order n is the exact integer ratio $4^n/C_n$ (exact integer arithmetic):

n	C_n	4^n	$4^n/C_n$
1	1	4	4.00
2	2	16	8.00
5	42	1,024	24.38
10	16,796	1,048,576	62.43
20	6,564,120,420	$\approx 1.10 \times 10^{12}$	167.50
30	$\approx 3.81 \times 10^{15}$	$\approx 1.15 \times 10^{18}$	302.21

Being $\sim \sqrt{\pi} n^{3/2}$, the improvement is polynomial: it does *not* move the exponential threshold. In an application with per-vertex activity ε , both bounds give the same radius requirement $4M^2\varepsilon < 1$ asymptotically; what changes is every finite-order coefficient, hence every truncation-error estimate at fixed order. Concretely, at $(d, \kappa_0) = (2, 16)$ (certified row: $M \in 1.0094221 \pm 4 \times$

10^{-8}), truncating the majorant series after order $n = 10$ leaves a tail whose leading coefficient is smaller by the factor 62.43 under the Catalan form — one and a half decimal digits of truncation error at no analytic cost. For the near-critical certified row $(3, 26)$, where $M \approx 4.9 \times 10^3$ makes every digit expensive, the same factor is the difference between a usable and an unusable finite-order bound. These consequences are paper-level arithmetic on top of the exact theorems.

7 Non-vacuity, machine-checked

The characteristic failure mode of formalized expansion bounds is hypotheses that no concrete geometry satisfies. Module `MarkedRootedClosure/NonVacuity.lean` therefore exhibits — as instance terms, not existentials — the hole family $\mathcal{H}^w$ in $d = 2$, $L = 8$ with two singleton holes at sup-distance 4, and proves by kernel decision (`decide`) that the holes are disjoint, edge-separated, and nonempty; the gate (1) at $\kappa_0 = 16$ is proved in Lean via $e^{16} > 2.7^{16} > 82944$ — the Lean-exact counterpart of the certified row $G \approx 0.00933$. Both polymer endpoints are then instantiated at $\mathcal{H}^w$ with the *strictly positive* weight $W_{32} = e^{-32(\rho+1)}$ (no zero-weight degeneracy):

```
nonvacuous_markedRootLeafGeometricBound, nonvacuous_targetPreservingWeightedTreeBound.
```

What this witness proves: the hypothesis set of Theorems 3.3–3.5 is satisfiable by a concrete geometry with positive weights, so the theorems have non-trivial instances and the bounds are statements about nonzero quantities. What it does not prove: that the instantiated bounds are sharp, or that the witness geometry models any physical large-field configuration; it is a consistency certificate, not an application.

8 Related work and novelty

Cluster expansions. The abstract polymer framework and its convergence criteria go back to Kotecký–Preiss [1], with the modern sharp form of the criterion due to Fernández–Procacci [2]; pedagogical accounts include Brydges [3]. Tree-graph inequalities dominating Ursell coefficients by spanning-tree counts originate with Penrose [4]; the family of tree and forest formulas is surveyed by Abdesselam–Rivasseau [5]. The upstream repository formalizes a Kotecký–Preiss criterion and a Mayer–Ursell expansion, and the majorants $S^\#, T^\#$ of Definition 2.5 are the standard objects of this tradition.

Holes and large fields. The specific geometry — hole-respecting polymers, skeletons, and the modified (Steiner) metric — is that of renormalization-group analyses of lattice gauge theory in the Balaban school [6], in the concrete form of Dimock’s expository series [7, 8, 9]; the hole-component families of Definition 2.2 follow [8, §3.1.2]. We are not aware of published *machine-checked* bounds for expansions in this geometry prior to the present programme.

The Catalan identity. The count of labelled rooted plane trees underlying Theorem 3.1 is classical combinatorics at the level of [10], and we make no novelty claim for the enumeration itself. What we have not found in the literature is its statement in exactly this normalized child-factorial form over spanning trees of K_{n+1} , its use to replace the 4^n shape budget inside an Ursell tree majorant with holes, or a formalization of either; readers aware of a prior source for the identity in this form are invited to point us to it.

What is new here. To our knowledge: (i) the formalized exact evaluation (Theorem 3.1) and its formalized corollary chain into hole-respecting leaf summation (Theorem 3.4); (ii) the machine-checked volume-free constant structure M^{2n+1} with the single-gate discipline (1); (iii)

the machine-checked non-vacuity witness; and (iv) the certified-constants companion with pre-registered bands and a documented registration failure. The mathematical ingredients are classical; the contribution is the verified integration.

Formalization context. The development builds on Mathlib [11]; the Catalan/plane-tree layer contributes upstream API (a `PlaneTree` companion submitted to Mathlib separately).

9 Reproducibility

Toolchain: Lean `leanprover/lean4:v4.29.0-rc6`; Mathlib pinned to commit

07642720480157414db592fa85b626dafb71355b

(lakefile and manifest agree); `python-flint` 0.9.0 on CPython 3.12.6; `tectonic` 0.15.0.

Theorem-to-artifact map (statuses as in Section 3; all Lean files in module `MarkedRootedClosure` unless noted):

Result	Lean name	File	Status
Thm. 3.1	<code>...TreeCatalanIdentity</code>	<code>PaperTheorems.lean</code>	exact
Thm. 3.2	<code>...TreeBound</code>	<code>PaperTheorems.lean</code>	exact
Thm. 3.3	<code>markedRootLeafGeometricBound</code>	<code>PaperTheorems.lean</code>	exact
Thm. 3.4	<code>markedRootCatalanBound</code>	<code>PaperTheorems.lean</code>	exact
Thm. 3.5	<code>targetPreservingWeightedTreeBound</code>	<code>PaperTheorems.lean</code>	exact
§7 witnesses	<code>nonvacuous_*</code>	<code>NonVacuity.lean</code>	exact
§5 table	—	<code>scripts/c1_admissibility_*</code>	certified
§6 ratios	—	integer arithmetic in text	exact
§6 consequences	—	—	paper-level

From a clean clone of the repository at the release revision:

```
lake exe cache get          # Mathlib oleans
lake build MarkedRootedClosure
lake env lean MarkedRootedClosure/Oracle.lean # axiom oracle
python scripts/c1_admissibility_arb.py      # certified table
tectonic -X compile \
  papers/c1-rooted-tree-majorants/c1_rooted_tree_majorants.tex
```

[SLOT: final release commit hash, tag, build job count, and the verbatim oracle output — inserted from the recorded logs at the release revision]

Pin-drift audit (measured): relative to the module’s original `archive/UPSTREAM.lock` (upstream commit `4e45246a...`), the blob of `KP/RootedLeafSummation.lean` is byte-identical at the build revision, while `RG/AppendixFSecondUrsellLeafSummation.lean` drifted additively (+144/ − 15 lines: the `treeShape` refactor and the Catalan route; the geometric endpoint’s statement is unchanged). The endpoints are compiled against the build revision, not the lock.

10 Conclusion

Five machine-checked theorems give the exact combinatorial value and the volume-free analytic envelope of rooted tree-graph majorants for polymer expansions with holes, with certified constants and a machine-checked consistency witness. The limitations are stated where they arise:

the signed-coefficient-to-majorant passage and the activity-series algebra are upstream respectively paper-level; the Catalan gain is polynomial and moves no radius; the witness certifies satisfiability, not sharpness. Reasonable next uses are (a) consuming Theorem 3.4 inside the upstream programme wherever the 4^n budget is currently paid, and (b) upstreaming the plane-tree API. Anything beyond that — in particular any continuum or gauge-theoretic significance — would require proved reductions that do not exist here, and none is claimed.

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